

ECO 209 Practice Questions

Name: _____

1. Explain the difference between flow and stock measures of economic, while identifying specific examples. Include a discussion (again on the basis of specific examples) of how such a distinction might be critical for comparing patterns of macroeconomic well-being and growth for different countries, regions and/or individuals.

Solution Flow measures account for economic activity over a specified period, typically representing a rate of change or flow per unit of time. Examples of flow measures include: output, investment, etc

Stock measures represent the accumulation of a particular economic variable at a specific point in time. Examples of stock measures include: Capital, debt, wealth

When comparing the economic performance of countries or regions, flow measures like GDP growth rates provide insights into the economic activity over certain period. However, stock measures such as wealth/capital per capita offer a more comprehensive view of economic well-being, it shows the accumulated assets and resources available to individuals or nations.

2. Determine the scale of return of the followings production function where Y is the output, N is the labour input and K is the capital input

(a) $Y = A(N + K)^\alpha$

(b) $Y = (N^\alpha + K^\alpha)^{\frac{1}{\alpha}}$

(c) $Y = AN^{0.3}K^{0.7}$

Solution For each of the following we will multiple the input by t times

(a) $Y = A(N + K)^\alpha$

$$Y = A(tN + tK)^\alpha = At^\alpha(N + K)^\alpha$$

It will depend on the α .

If $\alpha < 1$, the function is decreasing return to scale

If $\alpha = 1$, the function is constant return to scale

If $\alpha > 1$, the function is increasing return to scale

(b) $Y = (N^\alpha + K^\alpha)^{\frac{1}{\alpha}}$

$$Y = (N^\alpha + K^\alpha)^{\frac{1}{\alpha}} = ((tN)^\alpha + (tK)^\alpha)^{\frac{1}{\alpha}} = ((N^\alpha + K^\alpha)(t)^\alpha)^{\frac{1}{\alpha}} = t(N^\alpha + K^\alpha)^{\frac{1}{\alpha}}$$

This function is constant return to scale

(c) $Y = AN^{0.3}K^{0.7}$

$$Y = AN^{0.3}K^{0.7} = A(tN)^{0.3}(tK)^{0.7} = tAN^{0.3}K^{0.7}$$

This function is constant return to scale

3. List and explain three different ways that the government can finance their spending.

Solution

- (a) Increasing tax revenue: the government can finance its spending is by raising tax revenue. This involves imposing higher taxes on individuals, businesses, or certain economic activities to generate additional income for the government.
- (b) Increase money supply: the government finance its spending is by increasing the money supply. This typically involves the central bank purchasing government securities. The government can then use this newly created money to finance its expenditures.
- (c) Increase the amount of government borrowing: increasing the amount of government borrowing. This entails the government issuing bonds or taking out loans from domestic or international sources to raise funds

4. In a hypothetical economy, Eatonville, the output gap rise to 5% during the Information Technology boom. Many people believe that the multiplier was equal to 3. Given the equations that describe the economy:

$$C_t = \bar{a}_c \bar{Y}_t + \bar{x} \tilde{Y}_t \bar{Y}$$

$$I_t = \bar{a}_I \bar{Y}_t - \bar{b} (R_t - \bar{r}) \bar{Y}_t + 0.2 \tilde{Y} \bar{Y}_t$$

$$G_t = \bar{a}_G \bar{Y}_t$$

- (a) (2 pts) What is value of $\bar{x}$?
- (b) (1 pts) What would the percent change in government expenditure be to close this gap, assuming monetary policy is not being used?
- (c) (2 pts) Relying only on monetary policy and $\bar{b} = 0.5$, what is the required real interest rate to close the gap? Assumes that initially, R_t equals $\bar{r}$ (i.e., the boom is caused solely by a change in $\bar{a}$).
- (d) (1 pts) Why does the government and central bank want to cool down the economic boom? How does the policy help achieve this target?

Solution

(a)

$$3 = \frac{1}{1 - \bar{x} - 0.2}$$

$$\bar{x} = 0.466666666$$

(b)

$$\Delta a_G = -0.05/3 = -0.01667$$

(c) We want to solve for ΔR_t

$$\tilde{Y}_t = \frac{1}{1 - \bar{x} - 0.2} \left[\bar{a}_t - \bar{b} (R_t - \bar{r} + \Delta R_t) \right] = 0$$

$$\tilde{Y}_t = 3[-0.01667 - 0.5(0 - 0.03333)] = 0$$

(d) The government and central bank want to cool down the economy because they worry for the inflation. The policy would work because of the Phillip curve. When the economy is booming, the prices tend to increase as the firm need to hire more worker or increase costs to satisfy the demand.

5. Consider the consumption model discussed in Ch.16, the consumer has the following utility function:

$$U = \ln(c) + \beta \ln(c')$$

where c is the consumption in the current period, c' is the consumption in the future period and β is the discount factor. The consumer has an initial assets of f and can save or borrow f' amount at interest rate of r . In current period, the consumer receive y amount of income. In the future period, the consumer receives y' amount of income.

The government decided to implement a form of wealth tax. It will charge $\tau f'$ in the current period. The τ is the percentage that the government is going to tax on the total saving in current period (f').

- (a) (2 pts) Setup the budget constraint. Shows the current period budget constraint, future period budget constraint and the lifetime budget constraint.
- (b) (4 pts) Derive the Euler equation and compute the optimal c and c'
- (c) (4 pts) Now, the government decided to remove the wealth tax entirely (i.e., $\tau = 0$). Based on the equation derived, what is the effect on c and c' . What is the intuition behind it? Explain in 3-5 sentences.

Solution

(a)

$$c = y + f - f' - \tau f'$$

$$c' = y' + (1 + r)f'$$

$$c + \frac{c'(1 + \tau)}{1 + r} = y + \frac{y'(1 + \tau)}{1 + r} + f$$

(b) The utility function:

$$\ln \left(y + \frac{(1 + \tau)y'}{1 + r} + f - \frac{(1 + \tau)c'}{1 + r} \right) + \beta \ln(c')$$

The first order condition w.r.t. c' :

$$\left(\frac{1}{c} \right) \frac{1 + \tau}{1 + r} = \frac{\beta}{c'}$$

Rearrange:

$$\frac{c'}{c} = \beta \frac{1 + r}{1 + \tau}$$

Plug the Euler equation back to the budget constraint.

$$c + \frac{\beta^{\frac{1+r}{1+\tau}} c (1 + \tau)}{1 + r} = y + \frac{y'(1 + \tau)}{1 + r} + f$$

$$c + \beta c = y + \frac{y'(1 + \tau)}{1 + r} + f \rightarrow c = \left(\frac{1}{1 + \beta} \right) \left[y + \frac{y'(1 + \tau)}{1 + r} + f \right]$$

, and

$$c' = \beta \frac{1+r}{1+\tau} \left(\frac{1}{1+\beta} \right) \left[y + \frac{y'(1+\tau)}{1+r} + f \right]$$

- (c) If the $\tau = 0$, consumer would consume more of future consumption. It is because of the positive substitution effect and positive income effect. The consumer would have incentive to save more and consume more in the future. Meanwhile, the consume would reduce the current consumption, although consumer still has the positive income effect (less tax = higher income).

– The End –